BDA400 • DATA SCIENCE TOOLS AND TECHNIQUES

Technical Analysis Using R

Assignment 5 — Development Stage

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ASSIGNMENT VALUE	15%
PROJECT REPOSITORY	github.com/maljarad30-coder/TechnicalAnalysis
DATA PERIOD	October 31, 2025 – October 30, 2026

A reproducible implementation of nine technical analysis functions using base R, with stock filters and a moving-average trading rule.

Implementation Overview

Each indicator is implemented in a separate R script under **functions/**. The calculations use R core functions and do not call indicator libraries.

FUNCTION	PURPOSE / IMPLEMENTATION	OUTPUT
sma(data, period)	Equal-weight rolling mean over each complete window.	Numeric vector of length n – period + 1
ema(data, period)	Recursive exponential average; first value seeds the series.	Numeric vector of length n
macd(data, short, long, signal)	Short EMA minus long EMA; signal EMA and histogram are then derived.	List: MACD, signal, histogram
stdev(data)	Population standard deviation using the assignment's division by n.	Single numeric value
linreg(source, length, offset)	Least-squares slope and intercept for the selected trailing window.	List: slope, intercept, predictions
rsi(data, period)	Gain/loss changes with Wilder smoothing; values begin after a full period.	Numeric vector aligned to input; leading NA
stoch_rsi(data, period, k, d)	Rolling RSI normalization followed by %K and %D SMA smoothing.	List: aligned %K and %D vectors
crossover(arr1, arr2)	Detects upward and downward changes in relative position.	Labels: Up, Down, None
crossunder(arr1, arr2)	Detects when arr1 falls below arr2 after being at/above it.	Labels: True, False, None

Source files

The Assignment 2 CSV exports provide Date, Open, High, Low, Close, and Volume for AAPL, MSFT, and GOOGL. The old TTR-generated SMA20 column is ignored; Assignment 5 recalculates SMA with the base-R function in this project.

Filters and Trading Rule

The assignment description also calls for stock filtering and trading rules. These are included in **functions/10_filters_and_trading_rules.R** and applied by the runner.

Filter used in the example

The workflow selects rows from the latest 90 calendar days and keeps observations where the closing price is above the independently calculated SMA20. The filter function also accepts optional start/end dates and a minimum volume.

Trading rule

The example compares the 20-day and 50-day simple moving averages. An upward crossover changes the position to Long; a downward crossover changes it back to Flat. The event and position are included in each stock output.

Run instructions

01 Open the TechnicalAnalysis project in RStudio and set the working directory to the **Assignment5** folder.

02 Run **99_validate_functions.R** from the Console. It checks known examples, output lengths, and crossover signals. It should print “All Assignment 5 indicator checks passed.”

03 Run **00_run_assignment5.R** by selecting Source or entering `source("00_run_assignment5.R")`.

04 Review the printed summary and the generated files in **Assignment5/outputs/**. The runner writes one full technical-analysis CSV and one filtered CSV per ticker, plus a summary CSV.

> **Execution note: the source project data are already included in Assignment2/outputs. This Assignment 5 workflow does not download market data or require external R packages.**

Project files

Nine separate indicator scripts, a filters/trading-rules script, a runner, a validation script, and a README are included in the Assignment5 repository folder.

Technical Notes and Submission

Input data and scope

The included daily OHLCV CSV files contain 251 observations per ticker from October 31, 2025 through October 30, 2026. The original Assignment2 folder is retained; Assignment5 reads its CSV data and writes all new results into its own outputs folder.

Calculation conventions

• SMA returns only complete windows; leading NA values are added when aligning indicator results to trading dates.

• EMA is seeded with the first data point, following the supplied pseudocode. MACD uses 12-, 26-, and 9-period EMAs.

• RSI uses Wilder smoothing. A zero-loss period returns 100, a zero-gain period returns 0, and a completely unchanged period returns 50 to avoid undefined values.

• StochRSI uses a rolling RSI high/low range over the RSI period, then SMA smoothing for %K and %D. If the range is flat, the raw StochRSI value is set to 0.

• The crossover and crossunder outputs follow the labels in the supplied pseudocode, including None at the first observation.

Submission checklist

• Upload this PDF to the LMS as **Aljarad_Mohammad_CA_BDA400_A05.pdf**, following the filename pattern shown on the assignment page.

• Keep all Assignment 5 R scripts and README in the separate Assignment5 folder in the project repository.

• Run the validation and analysis scripts in RStudio, review the generated outputs, then commit the Assignment5 folder to the repository so the cover-page link shows the submitted work.

Prepared for BDA400 — Assignment 5